FORM 5: Z+F IMAGER 5016 SCAN STATION REGISTER & INTEGRITY SHEET

Terrestrial Laser Scanning Observation Log & Station Integrity Proforma | Balala Heritage Project | Target Layer: `tls_stations`

GEODETIC DATUM & PROJECTION: GDA2020 / MGA Zone 56 (EPSG:7856) | **Vertical Control:** AHD via AUSGeoid2020 | **STANDARDISED ID FORMATS:** Station: `BAL_ST001` | Targets: `TRG_SPH01`, `TRG_CHK01` | **GeoPackage:** `balala_master.gpkg`

SECTION 1 HEADER METADATA, CARRIER-WAVE ATMOSPHERIC CORRECTION & INSTRUMENT CONFIG

Project ID: Balala Heritage Project ..... **Survey Date (DD/MM/YYYY):** ..... **Laser Scanner Operator(s):** ..... **Weather & Lighting Conditions:** .....

ATMOSPHERIC INPUTS (EDM CORRECTION):
Mandatory for carrier-wave refractive index velocity model

temp_celsius (°C): ..... **pressure_hpa (hPa):** ..... **humidity_pct (%):** ..... **Computed ppm corr:** .....

Model: Z+F IMAGER 5016 ..... **Scanner Serial No.:** ..... **Firmware Version:** ..... **Raw File Prefix (e.g. BAL26_):** ..... **Structure / Sector Context (e.g. Woolshed East):** .....

SECTION 2 SCAN STATION OBSERVATION LOG (1:1 DIRECT GEOPACKAGE tls_stations SCHEMA)

Observation Protocol: *Record station immediately upon setting tripod. Measure HI (`hi_m`) to 3 decimal places from ground mark to scanner tilt axis notch. Verify RTK occupied mark ID if centred over peg. Record all visible sphere and checkerboard target IDs. Check HDR Colour if internal camera panorama is enabled.*

station_id (PK: BAL_ST01)	time_start (hh:mm)	hi_m (m: 3 dec)	res_setting (L / N / H / SH)	quality_mode (1x / 2x / 4x / 8x)	hdr_color (Colour Y/ N)	occupied_mark_id (FK: rtk_points)	control_target_ids (e.g. SPH01, SPH02, CHK01)	raw_filename (e.g. BAL_ST001.zfs)	sha256 check	target_feature / Sector Description (Structural element, room, or landform)
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	
			☐L ☐N ☐H ☐SH	☐1 ☐2 ☐4 ☐8	☐Y ☐N				☐	

Z+F IMAGER 5016 Scanning Conventions & Controlled Vocabulary: **res_setting:** `Low` (3.1mm @ 10m, 13s) | `Normal` (1.6mm @ 10m, 26s) | `High` (0.8mm @ 10m, 52s) | `Super High` (0.4mm @ 10m, 1m45s).
quality_mode: `1x` (Standard / Exterior daylight) | `2x` (Clean / Balanced) | `4x` (High noise-reduction / Dark interiors) | `8x` (Ultra / Reflective metal / Long range).
hdr_color: Check `Y` if full 360° HDR internal panorama captured (adds 2.5 min). Check `N` if laser-only scan without RGB photogrammetry.

SECTION 3 DATA OFFLOAD RECONCILIATION, FIXITY (SHA-256) & DIGITAL CHAIN OF CUSTODY

Raw Project Folder / Media ID: ..... **Total Scan Count on Card:** ... **Total Raw Volume (GB):** ..... **Fixity: All SHA-256 Hashes Computed? [ ] YES** ..... **Backup Drive 1 (Primary Field NAS):** .....

GeoPackage Ingest Layer: tls_stations [ ] Verified ..... **Ingest Date (DD/MM/YYYY):** ..... **Lead Surveyor Signature:** ..... **Archive Custodian Sign-off:** .....

Project: Balala Heritage Project **Survey Date:** ..... **Structure / Sector Context:** ..... **Instrument: Z+F 5016** ..... **Sheet No.:** .....

SECTION 4 STATION NETWORK, TARGET CONSTELLATION & SITE GEOMETRY SKETCH AREA (10×10 MM GRID)

N

SKETCH LEGEND: ▲ Scanner Station (`BAL_ST01`) | ● 150 mm Target Sphere (`TRG_SPH01`) | ■ Tile (`TRG_CHK01`) | **[X]** Peg (`BM_PEG01`) | –> Sighting Vector | ═ Wall / Footing | ▒ Obscured Zone

GRID SCALE (10mm sq): ☐ 1sq = 1m ☐ 1sq = 2m ☐ 1sq = 5m

SECTION 5 CONTROL TARGET REGISTER & RTK GNSS TIE TABLE (1:1 GEOPACKAGE tls_targets SCHEMA)

Target Sighting Protocol: *Minimum 3 non-collinear targets visible between adjacent scan stations (ideally 4-5). Disperse targets across range and elevation. Keep incidence angle to targets < 45°. Do not disturb control targets until all connecting station scans are completed.*

target_id (PK: e.g. TRG_SPH01)	target_type (150 mm sphere, tile, peg)	rtk_tie_id (FK: rtk_points.point_id)	est_distance_m (Dist to Scanner m)	mounting_ht_m (Target Centre HI m)	Station Sightings, Visibility & Occlusion Notes (e.g. Visible from ST01, ST02, ST03; tripod mounted)
	☐ 150mm Sph ☐ Tile ☐ Peg				
	☐ 150mm Sph ☐ Tile ☐ Peg				
	☐ 150mm Sph ☐ Tile ☐ Peg				
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	☐ 150mm Sph ☐ Tile ☐ Peg				

MANDATORY Z+F 5016 FIELD INTEGRITY RULES:
1. **Target Geometry:** Never place all targets in a single plane or straight line. Ensure spatial distribution in X, Y, and Z. Maintain target ranges between 2.5 m and 25 m for 150 mm spheres.
2. **Carrier-Wave Atmospheric Integrity:** Temperature and pressure must be re-recorded if ambient conditions shift > 3°C or > 5 hPa during the day to prevent EDM scale drift.
3. **Fixity Sign-Off:** Compute SHA-256 hashes of all `.zfs` raw scans upon memory card offload to ensure uncompromised digital chain of custody for archival deposit.